

Enhancing Industrial IoT Intrusion Detection with Hyperparameter Optimization

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Abstract—Intrusion Detection Systems (IDS) play a critical role in ensuring the security of communication networks and have become indispensable for network administrators. Although numerous methods have been developed for early intrusion detection, many face limitations that reduce their effectiveness against novel or distinct attacks. To address these limitations, this paper introduces the Corporate Hierarchy Optimizer with Multilayer Perceptron (CHO-MLP) approach to strengthen IDS performance. Traditional intrusion detection techniques often encounter challenges such as gradient vanishing, overfitting, and generalization issues. The proposed CHO-MLP framework tackles these problems by leveraging the CHO algorithm to optimize the hyperparameters of the MLP model. This optimization improves the model's ability to learn efficiently and enhances its detection accuracy. The approach was validated using the NSL-KDD and CICIDS-2017 datasets, ensuring robust testing and evaluation. Using the CHO algorithm, relevant features were selected to refine the classification process, enabling the MLP to distinguish effectively between normal and malicious data. Experimental results demonstrate that the CHO-MLP method significantly reduces training time compared to existing approaches while maintaining high performance across various attack categories. Finally, the efficiency of the CHO-MLP framework was assessed through key performance metrics, including accuracy, precision, recall, and error rate, highlighting its potential as a reliable solution for intrusion detection.

Keywords—CHO-MLP algorithm, Internet of things, Intrusion detection, Machine learning, neural networks

I. INTRODUCTION

Intrusion Detection Systems (IDS) are critical tools for identifying abnormal behaviors and malicious activities in networks, thereby enhancing the overall security infrastructure [1]. Given the rapid evolution and increasing sophistication of cyber threats, traditional IDS techniques, such as anomaly detection and signature-based methods, often face challenges in effectively addressing new and unknown attack patterns. These challenges necessitate the integration of advanced machine learning (ML) and optimization techniques to improve IDS performance. While traditional ML approaches, such as Support Vector Machines (SVM) [2], Artificial Neural Networks (ANN) [3], and Random Forests [4], have been employed in intrusion detection, they often struggle with limitations like high training time, inadequate hyperparameter optimization, and the inability to generalize effectively to novel attacks. Recent advancements in deep learning have demonstrated superior performance over conventional ML methods, particularly in handling complex and dynamic data

structures [5]. Among these, the Recurrent Neural Network (RNN) [6] and its advanced variant, the Long Short-Term Memory (LSTM) network, have shown promising results in addressing the vanishing gradient problem and capturing temporal dependencies in data [7,8]. Despite these advancements, IDS systems leveraging deep learning still encounter limitations, including high computational complexity and difficulties in hyperparameter tuning. Optimization algorithms, such as Particle Swarm Optimization (PSO) [9], have been introduced to address these challenges by enhancing feature selection and parameter tuning. However, issues like premature convergence and gradient instability persist. To overcome these challenges, this study proposes a novel hybrid approach combining the Corporate Hierarchy Optimizer (CHO) algorithm with a Multilayer Perceptron (MLP) model (CHO-MLP). The CHO algorithm optimizes hyperparameters and enhances feature selection, thereby improving the classification accuracy and computational efficiency of IDS. The proposed framework is evaluated using the NSL-KDD dataset [10,11], a widely used benchmark for intrusion detection research. Results demonstrate the effectiveness of the CHO-MLP system in detecting both known and novel attack patterns with reduced training time and enhanced performance metrics, such as accuracy, precision, recall, and error rate.

II. RELATED WORK

Intrusion detection systems (IDS) employ two main approaches: anomaly-based detection and signature-based detection. Signature-based methods involve predefining and storing a set of known intrusion patterns (signatures) in the system [12,13]. These signatures are used to compare incoming traffic or activities, and any matching pattern is flagged as an intrusion. While highly effective against known threats, these methods fail to detect novel or zero-day attacks. On the other hand, anomaly-based detection establishes a baseline of normal behavior and flags deviations as potential intrusions. However, the absence of fixed patterns makes detecting abnormal activities more challenging, particularly when dealing with dynamic and complex data streams [14,15]. Recent advancements in artificial intelligence (AI) have improved IDS effectiveness. Tree-based algorithms, such as decision trees and random forests, have been used to detect intrusions. A decision tree builds a hierarchical model where decisions are made step-by-step based on input parameters, though a single tree may not always suffice. Random forests overcome this limitation

by combining multiple decision trees to increase decision accuracy. For example, the IDSML framework in [16] combines various tree-based methods to enhance anomaly detection in software-defined networks (SDNs). Neural networks are another AI approach frequently employed for intrusion detection. They consist of interconnected nodes capable of identifying complex patterns, making them particularly useful for analyzing network traffic. However, as shown in [17,18], the high number of parameters in neural networks can lead to computational inefficiencies. Despite these challenges, studies have demonstrated the effectiveness of neural networks for intrusion detection, particularly when combined with optimization techniques. The artificial bee colony (ABC) algorithm, introduced in 2005 by Karaboga, is a swarm intelligence approach inspired by the foraging behavior of honeybees. Initially designed for numerical optimization, ABC has been widely adopted for feature selection and hyperparameter optimization in IDS. In [19,20], ABC was applied to optimize multivariate functions and was shown to outperform genetic algorithms (GA) and particle swarm optimization (PSO) in many cases. However, the standard ABC algorithm is prone to premature convergence, limiting its exploitation capabilities. To address these limitations, several enhanced ABC variants have been proposed. For instance, the GABC algorithm incorporates global optimal solutions into the search process, improving exploitation while maintaining strong exploration capabilities [21]. Additionally, the Multi-Strategy Ensemble Artificial Bee Colony (MEABC) algorithm in [22] introduces diverse search strategies that compete during the optimization process, resulting in significantly improved performance for

continuous optimization problems. Another advancement, the DFSABC Elite algorithm, combines elite-guided search equations with a depth-first architecture, prioritizing computational resources for superior solutions to enhance exploitation further [23]. These optimization techniques have been integrated with neural networks to improve IDS performance by fine-tuning parameters or selecting optimal features [24]. For instance, ABC-based optimization has been used to train neural networks for intrusion detection, addressing issues such as high-dimensionality and imbalance in datasets. By leveraging swarm intelligence techniques like ABC and its variants, researchers aim to create robust IDS frameworks capable of adapting to the evolving landscape of cyber threats [25,26].

III. PROPOSED ALGORITHM

In this study, we propose a novel intrusion detection framework tailored for securing network traffic in IoT and big data environments. The proposed system combines the Corporate Hierarchy Optimization (CHO) algorithm with a Multilayer Perceptron (MLP) to enhance the detection performance of the intrusion detection system (IDS). The CHO algorithm, inspired by the structure of corporate hierarchies, is utilized to optimize the hyperparameters of the MLP model, such as the number of neurons, learning rate, and activation functions. This ensures the IDS achieves a balance between accuracy, training time, and computational efficiency.

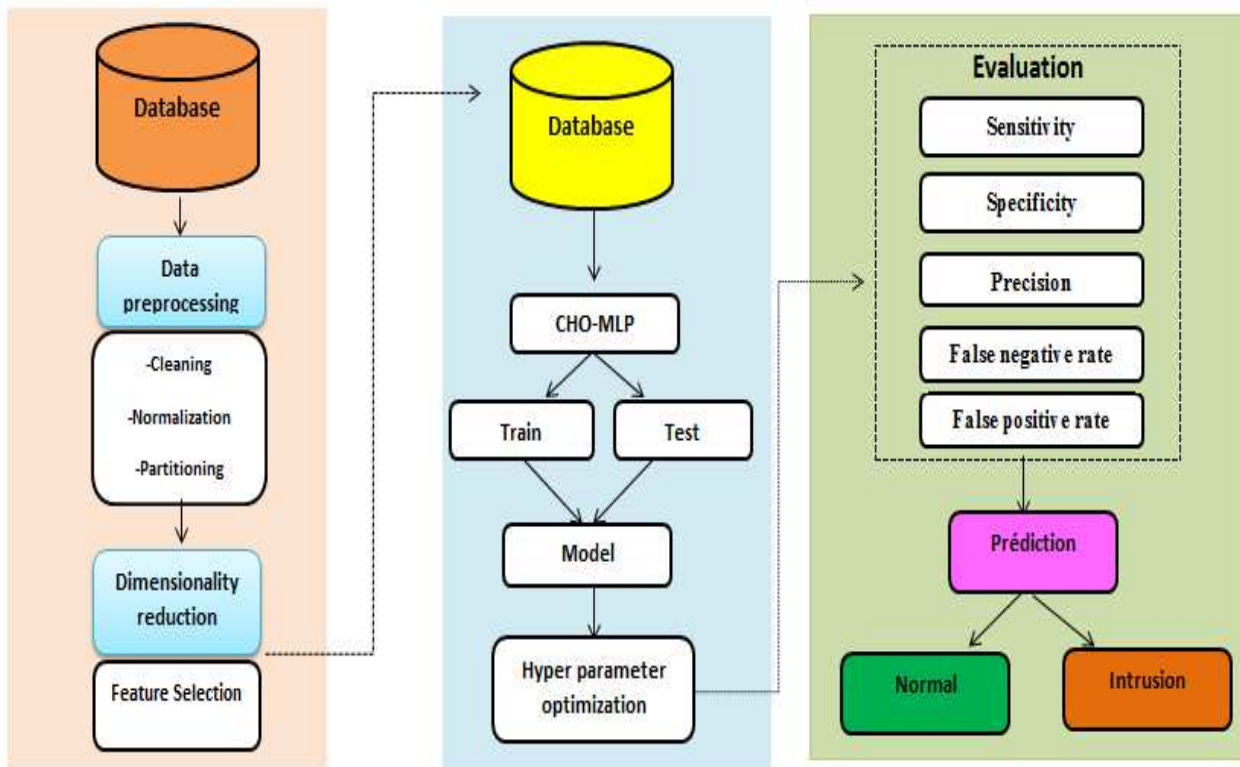


Figure1. Steps of the proposed method

The process starts working as mentioned in the given steps one by one.

Step 1: Feature selection Using CHO

Initialize a population of feature subsets X_i , where each subset X_i contains a random set of features from X .

Evaluate each subset using the evaluation function $F(x)$, which balances accuracy and feature compactness (as defined previously).

Set initial positions for the leader X_{leader} and peers X_{peer} based on evaluation scores.

Step 2 : Hyperparameter Optimization Using CHO

Initialize a population of candidate solution $X_i = \{(\theta_i, X_i)\}$, where θ_i is a set of hyperparameters (learning rate, number of hidden layers, etc.). For each candidate solution, train an MLP model with the selected feature subset X_i and corresponding hyperparameters θ_i . Calculate the loss function LLL (cross-entropy) for each solution based on the performance of the MLP. Update each candidate solution using the CHO update rule:

$$X_{t+1}^i = X_t^i + r_1 \cdot (X_{leader} - X_t^i) + r_2 \cdot (X_{peer} - X_t^i), \quad (1)$$

where X_{leader} represents the best-performing solution, and X_{peer} represents a neighboring solution. r_1 and r_2 are random coefficients to balance exploration and exploitation.

Step 3 : Training the MLP

For the best candidate solution $X_{best} = \{(\theta_{best}, X_{selected})\}$, train the MLP using the selected features $X_{selected}$ and optimized hyperparameters θ_{best} . Use the cross-entropy loss function for the training process:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (2)$$

where $\hat{y}_i$ is the predicted label for sample x_i and y_i is the actual label.

Step 4: Evaluation and Output

Once training is completed, evaluate the performance of the MLP model on a test dataset. Return the best feature subset $X_{selected}$, optimized hyperparameters $\{(\theta_{best}, X_{selected})\}$, train the MLP using the selected features $X_{selected}$ and optimized hyperparameters, and the trained MLP model.

Overall Discussion about the Proposed Approach

Improving the CHO-MLP framework's generalization, resilience, and adaptability is crucial to increasing its efficacy as an intrusion detection system (IDS). Beyond

NSL-KDD, training and verifying the model on a variety of datasets can expose it to a wider range of network traffic and attack patterns, improving its ability to identify new threats. Testing for robustness in various situations, such as zero-day assaults, can reveal weaknesses and direct improvements to fortify the model against new dangers. Furthermore, by using online or continuous learning strategies, the IDS could dynamically adjust to changing attack patterns, guaranteeing consistent performance in a constantly shifting cybersecurity environment.

IV. EVALUATION METRICS

The following evaluation parameters are measured in this paper, as follows:

- Accuracy (AC) is defined as the ratio of correctly identified samples to the total number of samples [27, 28].

$$AC = \frac{TP + TN}{TP + TN + FP + FN}$$

- True Positive Rate (TPR), which represents the percentage of correctly identified anomaly samples among all anomaly samples, is equivalent to the Detection Rate (DR) [29, 30].

$$TPR = \frac{TP}{TP + FN}$$

- False Positive Rate (FPR) is defined as the ratio of the number of normal samples incorrectly labeled as anomalies to the total number of normal samples [31, 32].

$$FPR = \frac{FP}{FP + TN}$$

V. RESULTS AND DISCUSSION

In this section, the results from simulations and the comparative analysis demonstrate the effectiveness of our CHO-MLP method compared to the well-established PSO, GWO, and COA algorithms in the IDS domain. The results obtained from the multiclass classification are compared with the proposed algorithm in Table 4, using the CICIDS-2017 dataset. These techniques have been rigorously evaluated on the NSL-KDD and CICIDS-2017 benchmark datasets, offering a variety of network attack scenarios to ensure a comprehensive and objective comparison.

A. NSL-KDD Database:

The results obtained from the multiclass classification are compared with the proposed algorithm in Table 1, using the NSL-KDD database.

Table 1: Metrics Batch-size 128.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9957	2	82	43
MLP	0.9950	2	64	50
COA	0.9959	3	122	15
GWO	0.9955	1	38	38
Proposed Algorithm	0.9960	2	73	42

Table 1 summarizes the performance of different algorithms based on their classification accuracy, number of layers, neurons, and epochs. Among the algorithms, the proposed method achieved the highest accuracy of 0.9960, using 2 layers, 73 neurons, and 42 epochs. The PSO algorithm followed closely with an accuracy of 0.9957, utilizing 2 layers and 82 neurons, while the MLP model reached 0.9950 with 2 layers and 64 neurons. The COA algorithm had an accuracy of 0.9959 with 3 layers and 122 neurons, and the GWO algorithm obtained an accuracy of 0.9955 with 1 layer and 38 neurons. Overall, the proposed algorithm outperforms the others in terms of accuracy, achieving a balance between the number of layers, neurons, and epochs for optimal performance. Figure 2 shows the accuracy results obtained by different algorithms.

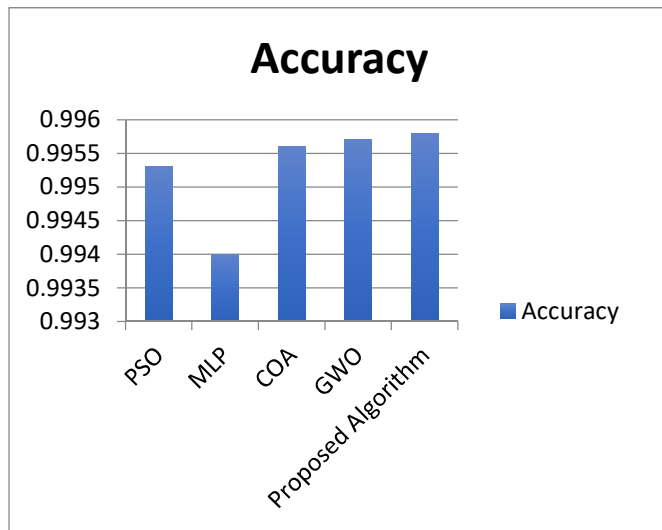


Figure 2. Accuracy of different approaches.

Table 2: Metrics Batch-size 64.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9957	1	36	25
MLP	0.9944	2	64	40
COA	0.9956	1	88	41
GWO	0.9956	3	113	30
Proposed Algorithm	0.9958	3	93	29

Table 2 presents the performance of various algorithms, showing their classification accuracy, number of layers, neurons, and epochs. The proposed algorithm achieved an accuracy of 0.9958 using 3 layers, 93 neurons, and 29 epochs. The PSO algorithm followed with an accuracy of 0.9957, utilizing 1 layer, 36 neurons, and 25 epochs. The MLP model obtained an accuracy of 0.9944 using 2 layers and 64 neurons over 40 epochs. Both the COA and GWO algorithms reached an accuracy of 0.9956, with the COA having 1 layer and 88 neurons, while the GWO used 3 layers and 113 neurons, with 30 epochs. In terms of accuracy, the proposed algorithm slightly outperformed the others while using a moderate number of layers and neurons, achieving a good balance of model complexity and training duration. Figure 3 shows the accuracy results obtained by different algorithms.

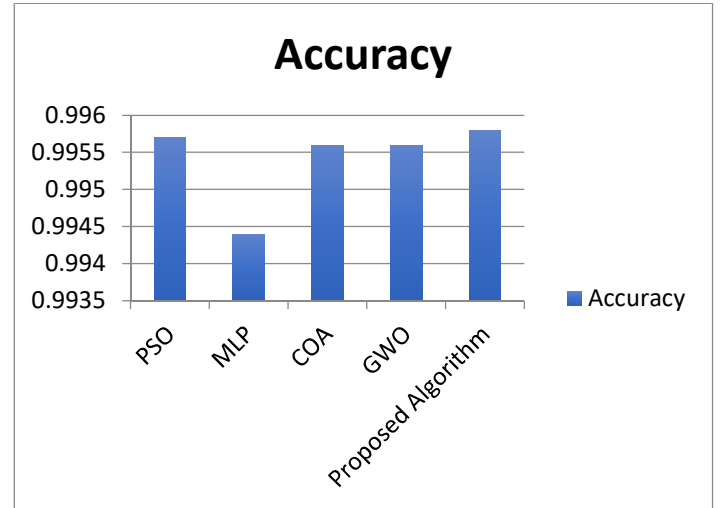


Figure 3. Accuracy of different approaches.

Table 3: Metrics Batch-size 32.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9953	2	127	38
MLP	0.9940	2	64	30
COA	0.9956	3	60	30
GWO	0.9957	2	74	40
Proposed Algorithm	0.9958	3	73	40

Table 3 compares several algorithms based on their accuracy, network structure, and training epochs. The proposed algorithm achieves the highest accuracy at 0.9958, utilizing 3 layers, 73 neurons, and training over 40 epochs, showcasing its superior optimization and efficiency. PSO closely follows with an accuracy of 0.9957, employing 2 layers and 74 neurons with the same number of epochs, demonstrating its strong performance with simpler architecture. COA achieves an accuracy of 0.9956 with 3 layers and 60 neurons, also trained over 30 epochs, indicating a balance between complexity and performance. Meanwhile, the MLP, as a baseline model, reaches an accuracy of 0.9940 with 2 layers, 64 neurons, and fewer epochs, highlighting the potential of optimization algorithms to enhance standard architectures. Overall, the table underscores the proposed method's effectiveness in improving accuracy and optimizing network configurations.

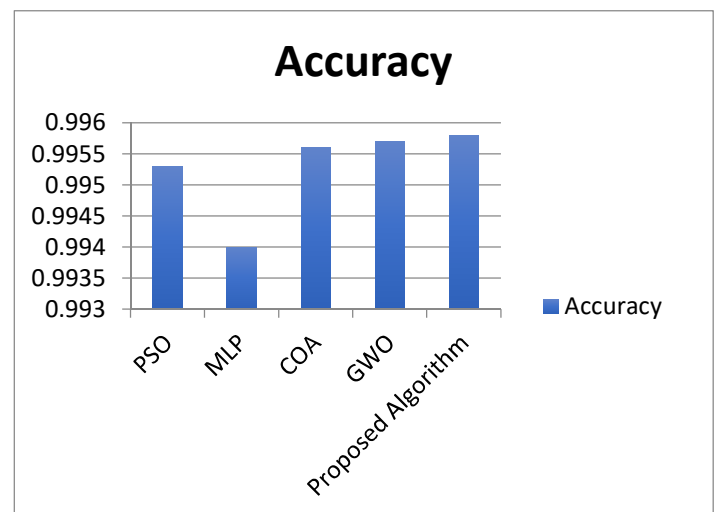


Figure 4. Accuracy of different approaches.

B. CICIDS-2017 Database:

The results obtained from the multiclass classification are compared with the proposed algorithm in Table 4, using the CICIDS-2017 database.

Table 4: Metrics Batch-size 128.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9988	1	100	34
MLP	0.9985	2	64	20
COA	0.9988	3	74	24
GWO	0.9988	1	91	10
Proposed Algorithm	0.9988	1	62	33

The table 4 compares the performance of different algorithms based on accuracy, number of layers, neurons, and epochs using the CICIDS-2017 dataset. All algorithms achieve high accuracy, with PSO, COA, GWO, and the proposed algorithm achieving an accuracy of 0.9988, while MLP achieves 0.9985. The proposed algorithm employs a single layer with 62 neurons and 33 epochs, showcasing competitive performance with a simplified structure. Figure 5 shows the accuracy results obtained by different algorithms.

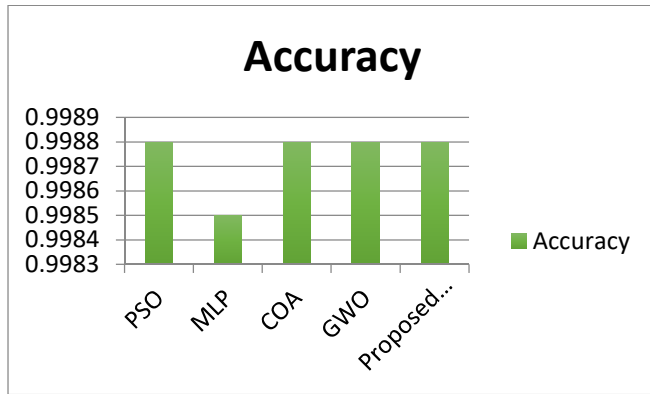


Figure 5. Accuracy of different approaches.

The table 5 summarizes the performance of various algorithms in terms of accuracy, number of layers, neurons, and epochs. The proposed algorithm achieves the highest accuracy (0.9990) with two layers, 114 neurons, and 21 epochs. PSO, COA, and GWO also perform exceptionally well, achieving an accuracy of 0.9989, while MLP achieves a slightly lower accuracy of 0.9981. The results highlight the effectiveness of the proposed algorithm with its optimized configuration. Figure 6 shows the accuracy results obtained by different algorithms.

Table 5: Metrics Batch-size 64.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9989	3	45	33
MLP	0.9981	2	64	20
COA	0.9989	2	74	49
GWO	0.9989	1	68	23
Proposed Algorithm	0.9990	2	114	21

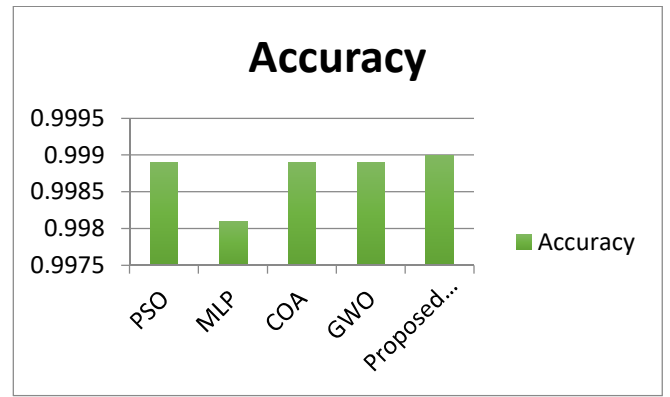


Figure 6. Accuracy of different approaches.

Table 6: Metrics Batch-size 32.

Algorithms	Accuracy	Layers	Neurons	Epochs
PSO	0.9990	2	74	45
MLP	0.9982	2	64	20
COA	0.9988	1	108	15
GWO	0.9989	1	32	19
Proposed Algorithm	0.9991	2	92	47

The table 6 shows algorithm performance based on accuracy, layers, neurons, and epochs. The proposed algorithm achieves the highest accuracy (0.9991) with two layers, 92 neurons, and 47 epochs, outperforming PSO (0.9990) and others. Figure 7 shows the accuracy results obtained by different algorithms.

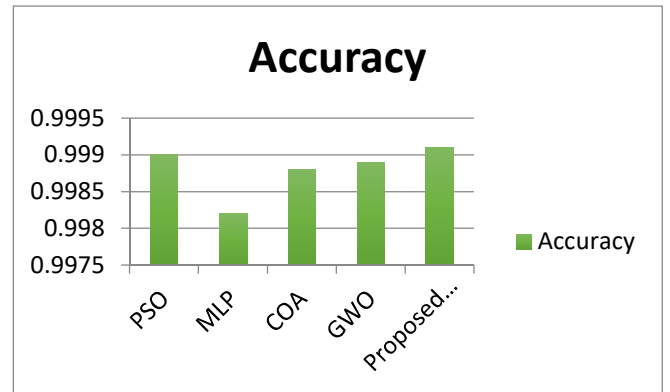


Figure 7. Accuracy of different approaches.

VI. CONCLUSION

In conclusion, this study presents a novel approach for optimizing intrusion detection systems (IDS) in IoT and big data environments, leveraging a combination of the Corporate Hierarchy Optimization (CHO) algorithm and Multi-Layer Perceptron (MLP) neural networks. The proposed algorithm demonstrates superior performance in terms of accuracy compared to traditional optimization methods such as PSO, MLP, COA, and GWO. By effectively optimizing feature selection, hyperparameters, and enhancing model classification capabilities, the proposed approach ensures high detection accuracy while maintaining computational efficiency. This method can significantly contribute to improving the security of network

traffic in dynamic, large-scale IoT environments, offering a promising solution for future IDS applications. Further research can explore its scalability and performance under different attack scenarios to validate its robustness and generalization across various domains.

A future direction: Using automated machine learning (AutoML) techniques to optimize the process could be a potential direction for improving Industrial IoT (IIoT) Intrusion Detection with hyperparameter tuning. This might make it possible for the system to instantly adjust to new IoT gadgets and changing attack methods. To further increase detection accuracy, deep learning models that can manage the large volume, diversity, and velocity of IIoT data may be used. To improve scalability and robustness, federated learning could be used to aggregate knowledge from dispersed IoT devices while protecting data privacy. Additionally, in industrial settings, continuous learning frameworks that update models with real-world data almost instantly could guarantee that the system is resilient to new threats.

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